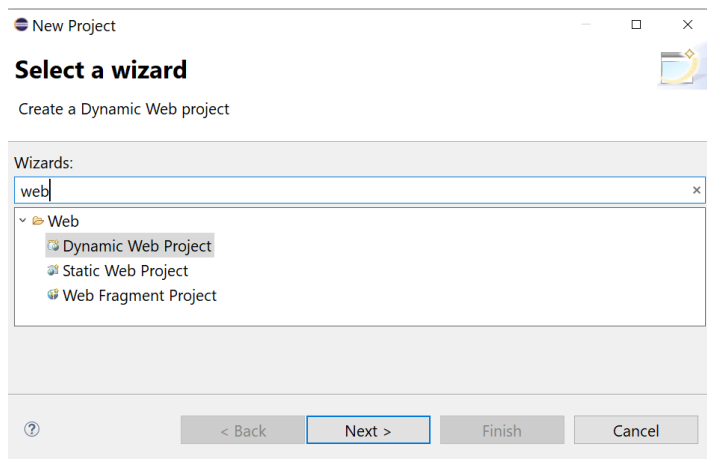

Creating XML Web application in Eclipse

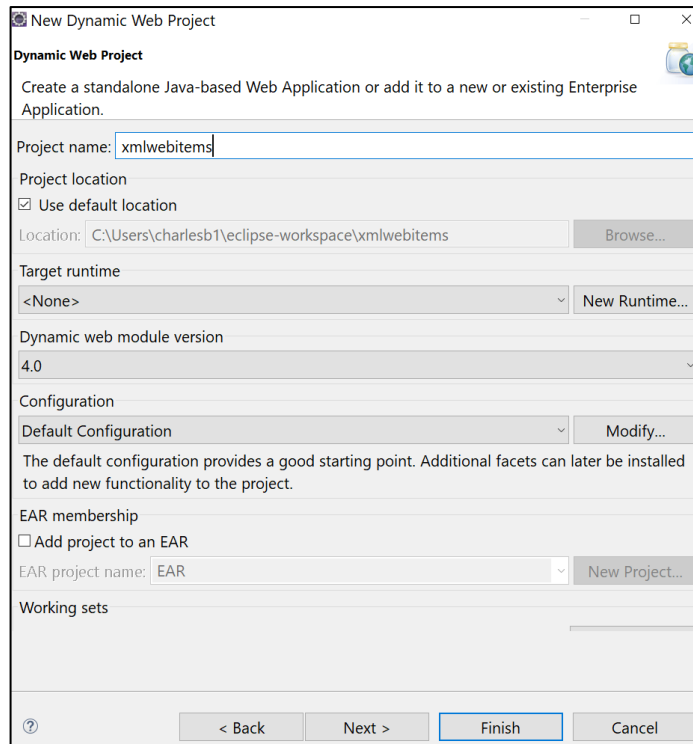
This tutorial illustrates how to create a web application to display XML data in a web page as shown in the table below

List of items		
Code	Name	Price
001	Bread	12.99
002	Rama	40.99
003	Coke	16.99

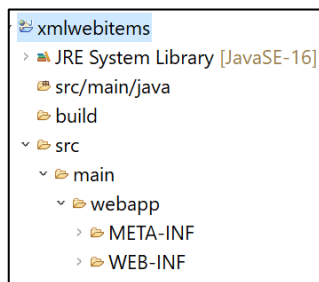
1. Create new Dynamic web project
 - a) Select new project
 - b) In the wizard, type Web
 - c) Select Dynamic Web pages and click Next



- d. In the window, type the name of the project (e.g., **xmlwebitems**), select the **Server** (under Target runtime) and click **Finish**

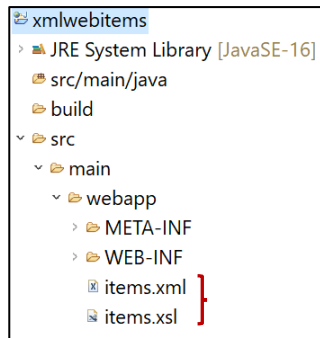


e. The Window below shows the folder structure of **xmlwebitems**



2. Create XML file in the **webapp** folde
 - a) Right-click **webapp**, select New and Other
 - b) In the wizard, type xml and select **XML, XML file** and click **Next**
 - c) In the new window, type the name and extension of the file (e.g. **items.xml**)
3. Create an XSL file in the **webapp**
 - a) Right-click **webapp**, select New and Other
 - b) In the wizard, search for **XLS, XLS file** and click **Next**
 - c) In the new window, type the name and extension of the file (e.g., **items.xsl**)

Below is the folder structure. Note: web resources are placed in the **webapp** folder



4. Write the XML code (e.g., **items.xml**) and validate (to validate, right-click file and select **Validate**)

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="items.xsl"?>

<Shop>
  <Item>
    <code>00001</code>
    <name>Coke</name>
    <price>24.99</price>
  </Item>
  <Item>
    <code>00002</code>
    <name>Bread</name>
    <price>18.99</price>
  </Item>
  <Item>
    <code>00003</code>
    <name>Milk</name>
    <price>15.00</price>
  </Item>
</Shop>
```

5. Write and validate the XLS code in **items.xsl** (to format the XML document). The data is presented in a table as shown below

List of items		
Code	Name	Price
001	Bread	12.99
002	Rama	40.99
003	Coke	16.99

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="2.0" xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:template match="/">
    <!-- TODO: Auto-generated template -->
    <h2>List of items</h2>
    <hr/>
    <table border="1">
      <tr>
        <th>Code</th>
        <th>Name</th>
        <th>Price</th>
```

```

        </tr>
        <xsl:for-each select="Shop/Item">
        <tr>
            <td><xsl:value-of select="code"/></td>
            <td><xsl:value-of select="name"/></td>
            <td><xsl:value-of select="price"/></td>
        </tr>
        </xsl:for-each>
    </table>
</xsl:template>
</xsl:stylesheet>

```

6. Link XSL document to the XML (as highlighted below)

```

<?xml-stylesheet type="text/xsl" href="items.xsl"?>
<Shop>
    <Item>
        <code>001</code>
        <name>Bread</name>
        <price>12.99</price>
    </Item>
    <Item>
        <code>002</code>
        <name>Rama</name>
        <price>40.99</price>
    </Item>
    <Item>
        <code>003</code>
        <name>Coke</name>
        <price>16.49</price>
    </Item>
</Shop>

```

7. Run the XML file

7.1. Without a server: Online tools for testing (e.g., TryIt Editor

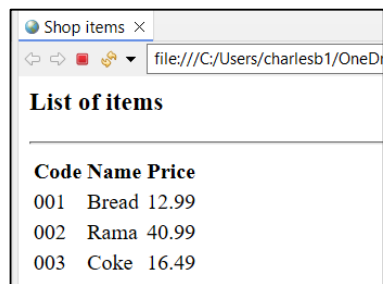
<https://www.w3schools.com/xml/tryxslt.asp?xmlfile=catalog&xsltfile=catalog>)

7.2. Without a server: In browsers **(not supported in most browsers for security reasons)**

7.3. Running on a server (e.g., Tomcat): Recommended **(will be demonstrated in class)**

For our purpose, we will test with Online tools.

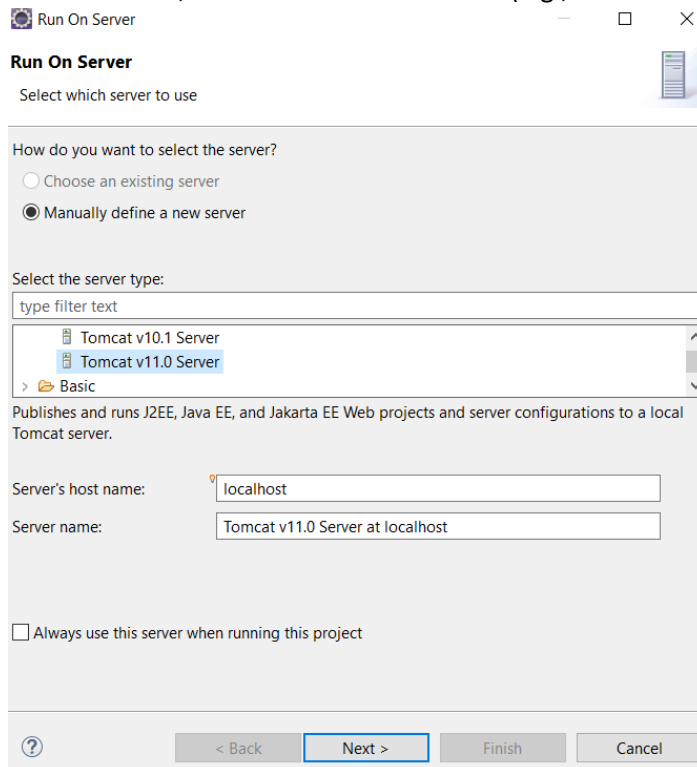
The data in the XML document is displayed in the browser as shown below.



Code	Name	Price
001	Bread	12.99
002	Rama	40.99
003	Coke	16.49

If you have installed a web server

- Right the xml file ((e.g., **items.xml**)
- Select **Run as -> Run on Server**
- In the window, select the **server installed** (e.g., Tomcat v11



Run On Server

Select which server to use

How do you want to select the server?

☐ Choose an existing server

☒ Manually define a new server

Select the server type:

type filter text

Tomcat v10.1 Server

Tomcat v11.0 Server

Basic

Publishes and runs J2EE, Java EE, and Jakarta EE Web projects and server configurations to a local Tomcat server.

Server's host name: localhost

Server name: Tomcat v11.0 Server at localhost

☐ Always use this server when running this project

< Back Next > Finish Cancel

1. Follow the instructions in Configuring Tomcat in Eclipse